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SYSTEM AND METHOD FOR VERIFYING IMAGE DATA IN REMOTE LOCATIONS

Abstract

A system and method for measuring an attribute of wildlife/marine life and confirming that measurement as legitimate by matching a digital code of the measurement device with a digital lock of the system is provided. A user may capture image data using a camera of a computing device, wherein the image data may contain measurement data that may be extracted and saved to a user's user profile. The system may determine if the data is legitimate based on a digital code contained within the image data and a digital lock of the system. The system may also create a digital fingerprint using a cryptographic hash function and data associated with the image data. The system may subsequently watermark a digital image with said digital fingerprint to create a watermarked digital image, wherein said digital image was created using said image data having said associated data.

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Background/Summary

CROSS REFERENCES [0001] This application is a continuation of a co-pending U.S. patent application Ser. No. 17/674,031 filed Feb. 17, 2022, which claims priority to U.S. Provisional Application No. 63/151,007, filed on Feb. 18, 2021, which application is incorporated herein by reference.

FIELD OF THE DISCLOSURE

[0002] The subject matter of the present disclosure refers generally to a system and method for verifying image data in remote locations.

BACKGROUND

[0003] Virtual sporting tournaments are becoming more popular due not only to convenience but also because it is necessary due to ever evolving public health concerns. The sport of fishing has been particularly active in adopting virtual sporting tournaments since catch and release tournaments are becoming more commonplace. However, virtual sporting tournaments do have some drawbacks. For instance, without the physical specimen in hand, additional steps must be taken in order to prevent cheating since the outcome of many of these tournaments are dependent on the word of the participants. If the attributes such as weight, length, location, time, etc. are not taken in a way that can alleviate all doubt, it is possible that fisherman participating in the event could manipulate the system in a way that gives them a decided advantage, and since many fishing tournaments award the top finishers with monetary rewards, developing methods to curtail cheating in these virtual sporting tournaments is a high priority.

[0004] Currently, the most common method to prevent cheating includes giving each participant a token that must be included in any image taken of caught marine life, wherein the token is only given to participants of the tournament in a time period prior to the start of the event that would make cheating less likely. The appearance of the token in the image is used to verify that the marine life was caught during the designated timeframe of the tournament. And though this method may help ensure that marine life was caught during the designated timeframe, it does not guarantee that a participant has not gained an advantage in some other fashion. For instance, the participant may be using a measuring instrument to measure the marine life that is not regulation. Or perhaps the participant used an image editing software to include the token in an image taken at a time prior to the start of the tournament. In addition to the limited cheating prevention potential of this method, transporting the token can be inconvenient to the participant. And if the participant leaves or loses the token, he may be disqualified from the event.

[0005] Accordingly, there is a need in the art for a system and method that allows participants in virtual sporting tournaments to remotely measure attributes of wildlife/marine life in a way that prevents many methods of cheating currently in usage today.

SUMMARY

[0006] A system and method for measuring an attribute of wildlife/marine life and confirming that measurement as legitimate by matching a digital code of the measurement device with a digital lock of the system is provided. In one aspect, the system allows a user to capture image data using a camera of a computing device, wherein the image data may contain measurement data that may be extracted and saved to a user's user profile. In another aspect, the system may determine if the image data is legitimate based on a digital code contained within the image data and a digital lock of the system. In yet another aspect, the system may create a digital fingerprint using a cryptographic hash function and data associated with the image data that may be used to create tagged image data. Generally, the system and method of the present disclosure allows users to remotely measure physical attributes of wildlife/marine life in a way that prevents said users from later manipulating data associated with said physical attributes.

[0007] The system generally comprises a computing device having a camera, light, control board operably connected to the light, power supply, processor operably connected to the computing device, and non-transitory computer-readable medium coupled to the processor and having instructions stored thereon. A user interface of the computing device may be used so that a user can login to a user profile so that image data obtained by the camera might be associated with a user profile of said user. Alternatively, a user identifier containing user data may be used by the system to associate image data with a particular user profile. In some embodiments, the user identifier may comprise a user identification (UID) pattern that may be captured within image data by the camera, wherein said UID pattern associates a particular user with a user profile of the system. In other embodiments, the user data may be contained within memory of a communication device of the user identifier, wherein said user data is transferred to the computing device prior to or during the capture of image data using the camera.

[0008] The light is configured to emit a pattern that associates the image with a particular tournament associated with the user profile of the user, which is captured within the image data and deciphered by the system. This pattern may be converted into a digital code that may match a digital lock of the system, which may be used to confirm data as legitimate or illegitimate after being captured by the various components of the system. The system may create digital fingerprints using data captured by the system, wherein said digital fingerprints may contain the information used to create it. These digital fingerprints are preferably encrypted in a way that prevents users of the system from deciphering the digital fingerprint without having the cryptographic hash function used to encrypt the digital fingerprint. The digital fingerprint may then be used to create tagged image data, wherein said tagged image data is preferably in the form of a digitally watermarked image or enhanced metadata image.

[0009] The foregoing summary has outlined some features of the system and method of the present disclosure so that those skilled in the pertinent art may better understand the detailed description that follows. Additional features that form the subject of the claims will be described hereinafter. Those skilled in the pertinent art should appreciate that they can readily utilize these features for designing or modifying other structures for carrying out the same purpose of the system and method disclosed herein. Those skilled in the pertinent art should also realize that such equivalent designs or modifications do not depart from the scope of the system and method of the present disclosure.

Description

DESCRIPTON OF THE DRAWINGS

[0010] These and other features, aspects, and advantages of the present disclosure will become better understood with regard to the following description, appended claims, and accompanying drawings where:

[0011] FIG. 1 is a diagram illustrating a system embodying features consistent with the principles of the present disclosure.

[0012] FIG. 2 is a diagram illustrating a system embodying features consistent with the principles of the present disclosure.

[0013] FIG. 3 is a diagram illustrating a system embodying features consistent with the principles of the present disclosure.

[0014] FIG. 4 is a diagram illustrating the manner in which individual access to data may be granted or limited based on user roles or administrator roles.

[0015] FIG. 5 is a flow chart illustrating certain preferred method steps that may be used by the system of the present disclosure.

[0016] FIG. 6 is a flow chart illustrating certain preferred method steps that may be used by the

system of the present disclosure.

[0017] FIG. 7 a diagram illustrating a system embodying features consistent with the principles of the present disclosure.

DETAILED DESCRIPTION

[0018] In the Summary above and in this Detailed Description, and the claims below, and in the accompanying drawings, reference is made to particular features, including method steps, of the invention. It is to be understood that the disclosure of the invention in this specification includes all possible combinations of such particular features. For example, where a particular feature is disclosed in the context of a particular aspect or embodiment of the invention, or a particular claim, that feature can also be used, to the extent possible, in combination with/or in the context of other particular aspects of the embodiments of the invention, and in the invention generally.

[0019] The term “comprises” and grammatical equivalents thereof are used herein to mean that other components, steps, etc. are optionally present. For example, a system “comprising” components A, B, and C can contain only components A, B, and C, or can contain not only components A, B, and C, but also one or more other components. Where reference is made herein to a method comprising two or more defined steps, the defined steps can be carried out in any order or simultaneously (except where the context excludes that possibility), and the method can include one or more other steps which are carried out before any of the defined steps, between two of the defined steps, or after all the defined steps (except where the context excludes that possibility). As will be evident from the disclosure provided below, the present invention satisfies the need for a system and method capable of managing equipment of a marine activity without human intervention, and thereby improving upon known systems currently employed within the art.

[0020] FIGS. 1-6 illustrate preferred embodiments of a system **100** and methods **500**, **600** designed to allow the remote measurement of wildlife/marine life **102** in a way such that it can be confirmed for accuracy. FIG. 1 illustrates a system **100** configured to collect image data **135D** and convert said image data **135D** into tagged image data **135E**, wherein said tagged image data **135E** contains a digital fingerprint **136** that confirms the user **105** and at least one of date, time, location, and physical attributes of marine life/wildlife. FIG. 2 illustrates a perspective view of the system **100** having weight sensors **240** and a slidable light sensor **215**. FIG. 3 illustrates how a user **105** may measure the length of a fish using the system **100** within an environment **300**. FIG. 4 illustrates permission levels **400** that may be utilized by the present system **100** for controlling access to user content **415**, **435**, **455**. FIG. 5 illustrates a method **500** for extracting measurement data **135C** from image data **135D** and subsequently verifying said measurement data **135C** using a digital code **114** and digital lock **113**. FIG. 6 illustrates a method **600** for encrypting measurement data **135C** and associating it with image data **135D** by way of a cryptographic hash function **130**. FIG. 7 illustrates a perspective view of the system **100** having a flexible tape **212** and a plurality of light sensors **215** in a line. It is understood that the various method steps associated with the methods of the present disclosure may be carried out as operations by the system **100** shown in FIG. 1.

[0021] As illustrated in FIG. 1, the system **100** of the present disclosure generally comprises a computing device **110** having a camera, light **235**, control board **210** operably connected to the light **235**, power supply, processor **115** operably connected to the computing device **110**, and non-transitory computer-readable medium **116** coupled to the processor **115** and having instructions stored thereon. The computing device **110** may comprise a user interface **111** that may be used to control the camera in order to capture image data **135D**, as illustrated in FIG. 3. In some preferred embodiments, a user interface **111** of the computing device **110** allows a user **105** to login to a user profile **135** so that image data **135D** obtained by the camera might be associated with a user profile **135** of said user **105**. In other preferred embodiments, the system **100** may include a user identifier **211** containing user data **135A** that is used by the system **100** to associate image data **135D** with a particular user profile **135**. In one embodiment, the user identifier **211** comprises a user identification (UID) pattern that may be captured within image data **135D** by the camera, wherein

said UID pattern associates a particular user with a user profile **135** of the system **100**. In another embodiment, the user data **135A** may be contained within memory of a communication device, wherein said user data **135A** is transferred to the computing device **110** prior to or during the capture of image data **135D** using the camera.

[0022] In some preferred embodiments, the light **235**, control board **210**, and user identifier **211** may be connected to a measurement device **107**, wherein said measurement device **107** comprises markings that when captured within image data **135D** can instruct the system **100** as to physical attributes of marine life/wildlife. Alternatively, the light **235**, control board **210**, and user identifier **211** may be secured to a card or hunting/fishing equipment, including, but not limited to, tackle boxes, fishing rods, fishing reels, marine vehicles, bows, and firearms. For instance, a compound bow may have an NFC device and light **235** embedded in the riser, wherein a computing device **110** of said may receive user data **135A** from the NFC device that associates said computing device **110** with a user profile **135** of said user **105**. Once the computing device **110** has been associated with the user profile **135**, the bow may be positioned near a deer shot by the user **105**, and the user **105** may then take an image of the deer and the bow. Because the computing device **110** is associated with a user profile **135** of the system **100**, image data **135D** taken with said computing device **110** will also be associated with this user profile **135**. The light **235** is configured to emit a pattern that associates the image with a particular tournament associated with the user profile **135** of the user **105**, which is captured within the image data **135D** and deciphered by the system **100**. In another preferred embodiment, the system **100** may further comprise at least one sensor **112**, which may be used to collect environmental data **135B** that may be used to assist the system **100** to confirm whether measurement data **135C** is legitimate. This environmental data **135B** may also be incorporated into the digital fingerprint **136** used by the system **100** to prevent the unwanted tampering of image data **135D** within the system **100**.

[0023] In another preferred embodiment, the system **100** may further comprise a database **125** operably connected to the computing device **110** having a plurality of user profiles **135** in which the various data of the system **100** may be stored. In yet another preferred embodiment, a server **120** may be operably connected to the database **125** and processor **115**, facilitating the transfer of information between the processor **115** and database **125**. In yet another preferred embodiment, at least one sensor **112** may be used to convey environmental data **135B** to the processor **115**. The various components of the system **100** of the present disclosure may be operably connected to one another via a wired or wireless connection via a communication device. The database **125** may be populated by a user **105** who uploads image data **135D** either directly to the database **125** or who inputs data into the user interface **111** of the computing device **110** that is subsequently transferred to the database **125**.

[0024] The system **100** is designed to measure an attribute of wildlife/marine life **102** and confirm that the measurement is legitimate by determining whether a digital code **114** of the measurement device **107** matches a digital lock **113** of the system **100**, wherein said digital code **114** is preferably a PIN. A PIN may be defined as a key having a predefined number of characters. The system **100** may compare the digital code **114** to a digital lock **113** of the system **100**, which may allow the system **100** to verify data as legitimate/illegitimate and/or unlock certain features of the system **100**, such as features of the user interface **111**. In some embodiments, the system **100** may extract measurement data **135C** and patterns from image data **135D** captured by the system **100**, wherein the patterns may be converted into one or more digital codes **114** and compared to one or more digital locks **113** associated with the user's **105** user profile **135**. In a preferred embodiment, patterns that the system **100** may convert into a digital lock **113** include, but are not limited to, QR codes, bar codes, light patterns, or any combination thereof.

[0025] Comparing the digital code **114** and digital lock **113** may instruct the system **100** as to whether or not the measurement device **107** is legitimate and/or where image data **135D** should be saved within a user profile **135** of a user **105**. In some embodiments, a matching digital code **114**

and digital lock **113** may also be required prior to the user **105** being allowed to capture image data **135D** using the system **100**. For instance, a digital code **114** contained within the user data **135A** of the user identifier **211** may be used to unlock a user interface **111** of a computing device **110** so that a user **105** may take image data **135D** with a camera of the computing device **110**. Preferably, the system **100** may encrypt data associated with the image data **135D** and subsequently create tagged image data **135E**, wherein tagged image data **135E** includes, but is not limited to, a digitally watermarked image, enhanced metadata image, and any combination thereof. The digitally watermarked image is preferably created and saved as imaged data is captured by the system **100**, thus preventing a user **105** from manipulating image data **135D** or any data associated with said image data **135D** after capture. The enhanced metadata image would have an encrypted digital fingerprint **136** saved within the metadata, wherein the encrypted digital fingerprint **136** contains information including, but not limited to, user data **135A**, date data, time data, geolocation data, tournament data, or any combination thereof.

[0026] Types of data that may be saved in the system **100** include, but are not limited to, measurement data **135C**, environmental data **135B**, image data **135D**, and user data **135A**, or any combination thereof. Measurement data **135C** may be defined as data containing information about a measurable attribute of wildlife/marine life **102**. Environmental data **135B** may be defined as data pertaining to a user's **105** environment. Image data **135D** may be defined as data containing pictures encoded as binary data, which preferably contain picture elements in the form of pixels. Image data **135D** may contain a single frame of reference or multiple frames of reference strung together to create a video. Image data **135D** may also contain metadata, including, but not limited to, audio data, date data, time data, camera setting data (aperture, shutter speed, focal length, metering mode, and iso speed), descriptions, copyright data, geospatial data, or any combination thereof. For instance, an image formatted in the exchangeable image format (Exif) may contain pixels, data and time information, geospatial data, and audio data. User data **135A** may be defined as information that may be used to identify a particular user **105**.

[0027] The processor **115** is configured to perform the operations disclosed herein based on instructions stored within the system **100**. The processor **115** may be any processor **115** or microprocessor **115** suitable for executing instructions. In some embodiments, the processor **115** may have a memory device therein or coupled thereto suitable for storing the data, content **415**, **435**, **455**, or other information or material disclosed herein. In some instances, the processor **115** may be a component of a larger computing device **110**. The processor **115** may process instructions for execution within computing devices **110**, including instructions stored in memory or on a storage device, to display graphical information for a graphical user interface **111** (GUI) on an external input/output device, such as a display. The processor **115** may provide for coordination of the other components of a computing device **110**, such as control of user interfaces **111**, applications run by a computing device **110**, and wireless communication by a communication device of the computing device **110**. A computing device **110** that may house the processor **115** therein may include, but are not limited to, laptops, desktops, workstations, personal digital assistants, servers **120**, mainframes, cellular telephones, tablet computers, or any other similar device. Accordingly, the inventive subject matter disclosed herein, in full or in part, may be implemented or utilized in devices including, but are not limited to, laptops, desktops, workstations, personal digital assistants, servers **120**, mainframes, cellular telephones, tablet computers, or any other similar device.

[0028] In a preferred embodiment, the programming instructions responsible for the operations carried out by the processor **115** are stored on a non-transitory computer-readable medium **116** ("CRM"), which may be coupled to the server **120**, as shown in FIG. **1**. Alternatively, the programming instructions may be stored or included within the processor **115**. Examples of non-transitory computer-readable mediums **116** include, but are not limited to, magnetic media such as hard disks, floppy disks, and magnetic tape; optical media such as CD ROM discs and DVDs;

magneto-optical media such as optical discs; and hardware devices that are specifically configured to store and perform programming instructions, such as read-only memory (ROM), random access memory (RAM), flash memory, and the like. In some embodiments, the programming instructions may be stored as modules within the non-transitory computer-readable medium **116**. The software instructions may be read into memory from another non-transitory computer-readable medium **116** or from another device. Alternatively, hardwired circuitry may be used in place of or in combination with software instructions to implement processes consistent with the principles of the invention. Thus, implementations consistent with the principles of the invention are not limited to any specific combination of hardware circuitry and software.

[0029] In an embodiment, the system **100** may further comprise a computing device **110** operably connected to the processor **115**. A computing device **110** may be implemented in a number of different forms, including, but not limited to, servers **120**, multipurpose computers, mobile computers, etc. For instance, a computing device **110** may be implemented in a multipurpose computer that acts as a personal computer for a user **105**, such as a laptop computer. For instance, components from a computing device **110** may be combined in a way such that a mobile computing device **110** is created, such as mobile phone. Additionally, a computing device **110** may be made up of a single computer or multiple computers working together over a network. For instance, a computing device **110** may be implemented as a single server **120** or as a group of servers **120** working together over and Local Area Network (LAN), such as a rack server **120** system **100**. Computing devices **110** may communicate via a wired or wireless connection. For instance, wireless communication may occur using a Bluetooth, Wi-Fi, or other such wireless communication device.

[0030] In a preferred embodiment, as illustrated in FIG. **1**, the computing device **110** is a mobile computing device **110**. Mobile computing devices **110** may include, but are not limited to, cellular telephones, smart phones, personal digital assistants, tablet computers, or other similar mobile computing devices **110**. In an embodiment, computing devices **110** may communicate audibly, meaning computing devices **110** may transmit and receive information via sound waves and convert the sound waves into digital information. For instance, a user **105** may instruct a user interface **111** of a computing device **110** with their voice to perform a certain action. The processor **115** may convert the sound waves of the user **105** into instructions, which the processor **115** may then carry out. Computing devices **110** may likewise generate audible sound for a user **105**, such as through an audio device. Such sound may include sound from voice telephone calls, recorded notes, voice messages, music files, etc. Audible sounds may also include sound generated by applications operating on a computing device **110**. For instance, an application running on a mobile computing device **110** may be configured in a way such that when a certain condition is met the application causes the mobile computing device **110** to output a sound. For instance, an application may be configured in a way such that an alarming sound is emitted via an audio device connected to the computing device **110** to indicate when a fishing tournament has ended. For instance, the processor **115** may receive a signal indicating that measurement data **135C** has been validated by the system **100**. The processor **115** may then convert this signal into an audio message that may be sent to an audio device to make the user **105** aware that the data has been validated.

[0031] As mentioned previously, the computing device **110** may further comprise a user interface **111**. A user interface **111** may be defined as a space where interactions between a user **105** and the system **100** may take place. In a preferred embodiment, the interactions may take place in a way such that a user **105** may control the operations of the system **100**. A user interface **111** may include, but is not limited to operating systems, command line user interfaces, conversational interfaces, web-based user interfaces, zooming user interfaces, touch screens, task-based user interfaces, touch user interfaces, text-based user interfaces, intelligent user interfaces, and graphical user interfaces, or any combination thereof. In some embodiments, the user interface **111** may be operably connected to back-end hardware, such as a server **120**, and/or software that separately

handles permission levels **400** of the various users **105**. The system **100** may present data of the user interface **111** to the user **105** via a display operably connected to the processor **115**. A display may be defined as an output device that communicates data that may include, but is not limited to, visual, auditory, cutaneous, kinesthetic, olfactory, and gustatory, or any combination thereof.

[0032] Information presented via a display may be referred to as a soft copy of the information because the information exists electronically and is presented for a temporary period of time. Information stored on the non-transitory computer-readable medium **116** may be referred to as the hard copy of the information. For instance, a display may present a soft copy of visual information via a liquid crystal display (LCD), wherein the hardcopy of the visual information is stored on a local hard drive. For instance, a display may present a soft copy of audio information via a speaker, wherein the hard copy of the audio information is stored on a flash drive. For instance, a display may present a soft copy of tactile information via a vibration device within the computing device **110**, wherein the hard copy of the tactile information is stored within a database **125**. Displays may include, but are not limited to, cathode ray tube monitors, LCD monitors, light emitting diode (LED) monitors, gas plasma monitors, screen readers, speech synthesizers, haptic suits, speakers, and scent generating devices, or any combination thereof.

[0033] In a preferred embodiment, the measurement device **107** comprises a platform **205**. The platform **205** preferably has markings **225** thereon that extend from a proximal end to a distal end of the platform **205**, which may be used to measure a length of wildlife/marine life **102**. In a preferred embodiment, the measurement device **107** further comprises a head piece **230** at the proximal end of the platform **205**. The markings **225** of the platform **205** preferably begin at the point in which the head piece **230** attaches to the platform **205**. In a preferred embodiment, the markings **225** allow a user **105** to measure the length of wildlife/marine life **102**. For instance, a user **105** may place one end of the wildlife/marine life **102** against said head piece **230** as a starting point to begin a measurement. In some preferred embodiments, the measurement device **107** may comprise a second head piece **231** at the distal end of said platform **205**. At least one of the head pieces **230**, **231** may be configured to move about the platform **205** in a way such that its position about the platform **205** may be changed, as illustrated in FIG. 2.

[0034] In another preferred embodiment, as illustrated in FIG. 7, the measurement device **107** may comprise a flexible tape **212** having markings thereon, which may be used to measure the length or girth of wildlife/marine life **102**. The flexible tape **212** preferably has markings **225** thereon that may be used by the system **100** to extract the length or girth of said wildlife/marine life **102** from said image data **135D**. The flexible tape **212** may extend perpendicularly to the axis on which the markings **225** pertaining to length extend. In yet another preferred embodiment, the measurement device **107** may further comprise at least one weight sensor **240**, wherein said at least one weight sensor **240** may be used to measure the weight of said wildlife/marine life **102**. A digital display **227** may be used to display the measured weight of the wildlife/marine life **102** to a user **105**. In embodiments of a system **100** comprising two head pieces **230**, **231**, wherein at least one of said head pieces is a sliding head piece **231**, wherein said sliding head piece **231** is slidably attached to rails **207** of said measurement device **107**. A distance measurer **220** secured to at least one of said head pieces **230**, **231** may be used to measure the distance between said head pieces **230**, **231**. In a preferred embodiment, an ultrasonic measurer or laser measurer may be used as the distance measurer **220**; however, other types of measurers may be used without departing from the inventive subject matter herein. A user **105** measuring the length of wildlife/marine life **102** using such a board may position one head piece **230**, **231** at a first end of the wildlife/marine life **102** and position the other head piece **230**, **231** at a second end of the wildlife/marine life **102**. The distance between the two head pieces **230**, **231** will coincide with the length of the wildlife, which may then be presented on the digital display **227**.

[0035] Other preferred embodiments of the system **100** may comprise at least one light sensor **215** configured to sense light in order to measure a length of marine life placed on the measurement

device **107**. In a preferred embodiment, as illustrated in FIG. 7, a plurality of light sensors **215** is placed in line about the length of the measurement device **107**. The system **100** may determine the length of the marine life based on the which light sensor **215** moving from a top light sensor **215** to a bottom light sensor **215** is the first to detect light and/or last to detect no light, wherein each sensor of the plurality of sensors located between said top light sensor **215** and said bottom light sensor **215** is associated with a particular length. Alternatively, the system **100** may determine the length of the marine life based on the which light sensor **215** moving from a bottom light sensor **215** to a top light sensor **215** is the last to detect light and/or first to detect no light. to slide about a track of the platform **205**. In another preferred embodiment, as illustrated in FIG. 2, a light sensor **215** attached to a sliding element may be used to measure the length of marine life placed on the measurement device **107**. A user **105** may slide the light sensor **215** up and down a track to which the sliding element is slidably secured in order to determine the point at which light is able to reach the light sensor **215**. When wildlife/marine life **102** is placed on the platform **205**, the earliest point in which light reaches the light sensor **215** may indicate the length of the wildlife/marine life **102**. In a preferred embodiment, the sliding element is a knob, but other sliding elements may be used without departing from the inventive subject matter described herein. In some preferred embodiments, a user **105** may manipulate a portion of the sliding element located on the back side of the platform, allowing a user **105** to move the light sensor **215** about the track. This may allow the user **105** to move the light sensor **215** without having to move any wildlife/marine life **102** placed on the face of the platform **205**.

[0036] A light **235** of the measurement device **107** may be used to indicate when sufficient light is reaching the light sensor **215** so that the length of marine life may be measured. For instance, a user **105** may place a redfish on a platform **205** of a measurement device **107** having a light sensor **215** and two lights **235**, wherein one light **235** is a red light and the other is a green light. When the redfish is initially placed on the measurement device **107**, the redfish may be covering up the light sensor **215** secured to a sliding element, and the measurement device **107** may indicate that light is not reaching the light sensor **215** by only powering the red light. The user **105** may slide the light sensor **215** about the track using the portion of the sliding element located on the back side of the platform **205** until the point at which light is reaching the light sensor **215**, and the measurement device **107** may then indicate that light is reaching the light sensor **215** by only powering the green light. In some preferred embodiments, markings **225** coinciding with the point at which the light sensor **215** is receiving light may indicate the length of the wildlife/marine life **102**. In other preferred embodiments, the system **100** may be configured to digitally provided a length of marine life based on the location of the sliding element about the length of the track.

[0037] In a preferred embodiments a light **235** configured to transmit a pattern may be attached to the measurement device **107**. Types of lights **235** that may be used by the system **100** to emit the pattern include, but are not limited to, light emitting diodes (LED), infrared (IR), fluorescent, or any combination thereof. The pattern emitted by the light **235** is determined by the control board **210** operably connected to the light **235**. In a preferred embodiment, the light **235** is attached to the platform **205**; however, the light **235** may be attached to at least one head piece **230**, **231** without departing from the subject matter described herein. In a preferred embodiment, the light **235** is attached to the measurement device **107** in a way such that both the pattern emitted by the light **235** and the markings **225** are visible when measuring wildlife/marine life **102**. Making both the light **235** and markings **225** visible when measuring wildlife/marine life **102** may allow a user **105** to capture both the markings **225** on the measurement device **107** and the pattern emitted by the light **235** when using the camera to capture image data **135D**.

[0038] Materials that may be used to construct the measurement device **107** include, but are not limited to, polymer, steel, aluminum, rubber, or any combination thereof. Some preferred embodiments of the measurement device **107** may be constructed in a way such that they are buoyant and float in water. For instance, the measurement device **107** may be completely

constructed of a material that has a density lower than that of water. For instance, the device may be partially constructed of a material having a density lower than that of water so at least a portion of the device floats when placed in water. For instance, the measurement device **107** may be constructed in a way that includes air pockets within the platform **205** of the measurement device **107**, which make the resulting measurement device **107** structure less dense than water despite the material used. Other preferred embodiments of the measurement device **107** may comprise a float. For instance, a measurement device **107** made of steel may have a float connected thereto, which may allow the measurement device **107** to float in water.

[0039] The control board **210** comprises at least one circuit and a microchip. Some embodiments of the control board **210** may further comprise a wireless communication device, such as Bluetooth, NFC, or an antenna. The wireless communication device may be used to transmit information to or receive information from a computing device **110**, including user data **135A** that the system **100** may use to associate a computing device (and therefore image data **135D** taken with a camera of said computing device) with a user profile **135**. The control board **210** may regulate the transfer of power to the various components of the measurement device **107** and control the pattern emitted by the light **235**. The microchip of the control board **210** comprises a microprocessor **115** and memory. The microprocessor **115** may be defined as a multipurpose, clock driven, register based, digital-integrated circuit which accepts binary data as input, processes it according to instructions stored in its memory, and provides results as output. The microprocessor **115** may receive a signal that instructs it to begin emitting the pattern via the light **235**.

[0040] In a preferred embodiment, the microprocessor **115** of the control board **210** receives said signal via a physical switch connected to the control board **210**. In a preferred embodiment, the switch is a button located on the measurement device **107** and operably connected to the control board **210**. When a user **105** interacts with the switch, a signal is sent to the microprocessor **115** of the control board **210**, which prompts the microprocessor **115** to relay the digital code **114** by way of a pattern via the light **235**. Alternatively, the microprocessor **115** may receive the signal to emit the pattern via a light **235** from a computing device **110** operably connected to the control board **210** via a wireless communication device. The microprocessor **115** may access the digital code **114** from memory of the control board **210** when prompted and convey the digital code **114** via the light **235**. In a preferred embodiment, the digital code **114** emitted by the light **235** associates image data **135D** with a particular event, such as a fishing/hunting tournament. However, the digital code **114** emitted by the light **235** may be used in other ways by the system **100** without departing from the inventive subject matter described herein. For instance, the digital code **114** emitted by the light **235** may serve as additional bit of information that may be used by the system **100** in conjunction with the user data **135A** of the user identifier **211** to associate image data **135D** with a particular user profile **135**.

[0041] In a preferred embodiment, pattern instructions **108** for emitting a pattern via the light **235** are stored in the memory of the control board **210**. For instance, the communication device may transmit pattern instructions **108** to the microprocessor **115** from the computing device **110** that cause the microprocessor **115** to retrieve pattern instructions **108** from the memory that instruct the processor **115** in how to carry out the transmission of said pattern using the light **235**. In some preferred embodiments, the system **100** may change the digital code **114** within the memory by pattern instructions **108** received via the wireless communication device. For instance, a user **105** may place a computing device **110** having an NFC device therein near the wireless communication device of the control board **210**. The computing device **110** may transmit pattern instructions **108** to the control board **210** that instruct the microprocessor **115** to alter the digital code **114** stored within the memory. By using the computing device **110** to change the digital code **114**, the digital code **114** may be associated with a particular event and subsequently used to verify that a user **105** has permission to participate in a particular event. For instance, the system **100** may comprise a plurality of events, wherein each event requires a unique digital lock **113** to participate therein. A

user **105** may enter the event using the user interface **111** of the system **100**, and the computing device **110** may then be used to update the control board **210** (via the wireless communication device) with a digital code **114** matching the digital lock **113** associated with the desired event. Therefore, in some embodiments, the digital code **114** of the system **100** may be used as a digital token to verify that a user **105** has entered image data **135D** into an event and/or is using a measurement device **107** authorized for said event.

[0042] Types of devices that may act as the wireless communication device include, but are not limited, to near field communication (NFC), Bluetooth, infrared (IR), radio-frequency communication (RFC), radio-frequency identification (RFID), and ANT+, or any combination thereof. In an embodiment, the wireless communication device may broadcast signals of more than one type. For instance, a control board **210** comprising an IR transmitter and RFID transmitter may broadcast IR signals and RFID signals. Alternatively, a wireless communication device may broadcast signals of only one type of signal. For instance, the control board **210** may be fitted with an equipment transmitter that may only transmit/broadcast RFC signals with a computing device **110**.

[0043] The computing device **110** may detect the signals broadcast by the wireless communication device. As illustrated in FIG. 1, the computing device **110** is operably connected to the processor **115**. In a preferred embodiment, the computing device **110** may be configured to detect a number of signal types that may be emitted by the control board **210**, including, but not limited to, NFC, Bluetooth, IR, RFC, RFID, and ANT+, or any combination thereof. For instance, a computing device **110** may be configured to detect Bluetooth, NFC, and IR signals broadcast by one or more control boards **210**. Alternatively, the computing device **110** may be configured to detect only one type signal emitted by a control board **210**. For instance, a computing device **110** may be to only detect Bluetooth signals. As such, control boards **210**, wireless communication devices, and computing devices **110** may be combined in multiple ways without departing from the inventive subject matter as described herein.

[0044] In another embodiment, at least one sensor **112** may be used to measure environmental conditions as image data **135D** is taken and transmit said environmental data **135B** to the processor **115**. An environmental condition may be defined as external conditions experienced by a user **105**. Types of sensors that may be used as an at least one sensor **112** include, but are not limited to, a microphone, turbidity sensor, thermometer, humidity sensor, passive infrared sensor, light sensor **215**, lightning sensor, wind transducer, compass, global positioning system (GPS), gyroscope, accelerometer, barometer, or any combination thereof. Therefore, the at least one sensor **112** may measure a variety of types of environmental data **135B** and transmit that data to the processor **115**. Once the processor **115** has received the environmental data **135B**, the processor **115** may save the environmental data **135B** within the user profile **135**. In some preferred embodiments, environmental data **135B** may be used to verify whether measurement data **135C** is legitimate. For instance, an event within the system **100** may have a defined geolocation range. A user **105** entered into that event may transmit geolocation data to the system **100** when capturing image data **135D**, and the system **100** may compare the geolocation data obtained with the geolocation range of the event to verify that a user **105** is within the specified geolocation range of said event.

[0045] Measurement data **135C**, environmental data **135B**, image data **135D**, and user data **135A** are preferably saved in user profiles **135** of the system **100**. In a preferred embodiment, the user profile **135** may be saved to the non-transitory computer-readable medium **116**. Alternatively, the user profile **135** may be saved to a database **125**. A user **105** may manually or automatically update a user profile **135** using a computing device **110**. In a preferred embodiment, the computing device **110** may comprise a user interface **111**, which a user **105** may use to manually or automatically update data within the user profile **135**. For instance, environmental data **135B** may be automatically uploaded to a user profile **135** when detected by the at least one sensor **112** of the system **100**. For instance, geospatial data may be automatically uploaded to a user profile **135** when

image data **135D** is captured by the computing device **110**. In another preferred embodiment, the user **105** may manually update data within user profiles **135**. This may be done through use of, but not limited to, a camera, scanning device, keyboard, mouse, voice recognition and/or biometric mechanisms, etc. For instance, a user **105** may manually capture image data **135D** using the camera of the computing device **110** prior to said data being automatically uploaded to a user profile **135**. For instance, a user **105** may use a keypad to manually input use data into a user interface **111** of the computing device **110**. Once a user **105** has manually captured/entered the data, the data may be transferred to a user profile **135** within the non-transitory computer-readable medium **116** and/or database **125** via the processor **115**. For instance, a user **105** may manually input environmental data **135B** into the user interface **111** of a computing device **110**, which may subsequently be obtained from the computing device **110** by the database **125** via a processor **115** operably connected to the computing device **110**. In a preferred embodiment, the image data **135D** is obtained by the database **125** from the computing device **110** via the processor **115** whenever a user **105** is in an area in which the computing device **110** has internet connectivity.

[0046] In some preferred embodiments, the system **100** may determine whether measurement data **135C** obtained from image data **135D** is legitimate or illegitimate by comparing the digital code **114**—obtained by the computing device **110** from the user identifier **211** and/or by transforming a pattern emitted by the light **235** (and captured within the image data **135D**)—to at least one digital lock **113** of the system **100**. Legitimate data may be defined as data that has been captured by a computing device **110** after said computing device **110** has been associated with user profile **135** via at least one digital code **114** that matches at least one digital lock **113** of said user profile **135**. In some preferred embodiments, measurement data **135C** may be both legitimate and illegitimate. For instance, the system **100** may tag measurement data **135C** as legitimate for measurement purposes but illegitimate as far as participation purposes for an event that a user **105** is not registered to compete in. This might occur in instances in which a. digital code **114** of a user identifier **211** matches a digital lock **113** of a user profile **135** but a pattern emitted by a light **235** does not match a digital lock **113** of a user profile **135** that would associate the image data **135D** with a particular event. In a preferred embodiment, when a user **105** captures image data **135D** using the camera of the computing device **110**, the control board **210** of the measurement device **107** may be prompted via a computer readable signal transmitted by the computing device **110** to cause the light **235** to emit a pattern that is captured within the image data **135D**. The system **100** may then extract from the image data **135D** both the measurement data **135C** and the pattern.

[0047] The pattern may be transformed into a digital code **114** based on instructions of the system **100**, and that digital code **114** may be compared to a digital lock **113** of the system **100**. In a preferred embodiment, the pattern emitted by the light **235** conveys binary the system **100** may extract and transform into the digital code **114**. However, other methods of encoding text characters may be used by the system **100** without departing from the inventive subject matter herein. For instance, the control board **210** may use Morse Code to convey a PIN to the computing device **110** via the light **235**. The system **100** may also use environmental data **135B** in combination with the PIN to verify measurement data **135C** as legitimate for a particular event. For instance, the system **100** may require that an encrypted digital code **114** and digital lock **113** not only be used to confirm that a user **105** is a legitimate participant in an event using an authorized measurement device **107** but may also require that geolocation data obtained while collecting image data **135D** confirm that the user **105** is in a geolocation authorized for the event.

[0048] In other preferred embodiments, the system **100** may use encryption techniques to alter the pattern emitted by the light **235**, digital code **114**, and/or digital lock **113**. For instance, an algorithm of the control board **210** and computing device **110** may cause the processors **115** of the system **100** to continuously alter the digital code **114** and/or digital lock **113** based on a clock. The digital code **114**, digital lock **113**, algorithm, and clock of a particular control board **210** and computing device **110** may be associated with a particular user profile **135**, which may allow the

system **100** to continuously alter the digital lock **113** of the system **100** to match the continuously changing digital code **114** of the control board **210**. Alternatively, the pattern of a particular board may remain constant and associated with a particular user profile **135**; however, an algorithm associated with a particular user's **105** user profile **135** may cause the computing device **110** to transform the pattern into a digital code **114** that the system **100** may then compare to a digital lock **113** of the user's **105** user profile **135**. This may be done via a clock and algorithm as previously described.

[0049] As used herein, a database **125** refers to a set of related data and the way it is organized. Access to this data is usually provided by a database **125** management system (DBMS) consisting of an integrated set of computer software that allows users **105** to interact with one or more databases **125** and provides access to all of the data contained in the database **125**. The DBMS provides various functions that allow entry, storage and retrieval of large quantities of information and provides ways to manage how that information is organized. Because of the close relationship between the database **125** and the DBMS, as used herein, the term database **125** refers to both a database **125** and DBMS.

[0050] As shown in FIG. **1**, the database **125** may be configured to store data relating to the measurement of wildlife/marine life **102** therein. The processor **115** may be operably connected to the database **125** via wired or wireless connection. In a preferred embodiment, the information, data, and/or content **415**, **435**, **455** associated with a measurement of wildlife/marine life **102** may be stored as measurement data **135C** within a user profile **135**. In a preferred embodiment, the database **125** may be configured to store a plurality of user profiles **135** therein and the various information, data, and/or content **415**, **435**, **455** tied to or associated with such profiles. The database **125** may be a relational database such that the measurement data **135C**, environmental data **135B**, image data **135D**, and user data **135A** within a use profile within the plurality user profiles **135** may be stored, at least in part, in one or more tables. Alternatively, the database **125** may be an object database such that the measurement data **135C**, environmental data **135B**, image data **135D**, and user data **135A** within a user profile **135** within the plurality of user profiles **135** may be stored, at least in part, as objects. In some instances, the database **125** may comprise a relational and/or object database and a server **120** dedicated solely to managing the content **415**, **435**, **455** assigned to user profiles **135** in the manner disclosed herein. Although the database **125** is represented as a single entity within FIG. **1**, it is understood that data, information, and/or content **415**, **435**, **455** stored within the database **125** or repository, as disclosed herein, may be stored within a plurality of databases **125** without departing from the inventive subject matter disclosed herein.

[0051] As mentioned previously, the system **100** may further comprise a server **120**. A server **120** may be a search server, a document indexing server, and general web server. Servers **120** may be separate entities performing different functions or similar functions. For instance, two or more servers **120** may be implemented to work as a single server **120** performing the same tasks. Alternatively, one server **120** may perform the functions of multiple servers **120**. For instance, a single server **120** may perform the tasks of a web server and an indexing server. Although represented as a single server **120** in FIG. **1**, it is understood that multiple servers **120** may be used to operably connect the processor **115** to the database **125** and/or the content **415**, **435**, **455** repositories. The processor **115** may be operably connected to the server **120** via wired or wireless connection.

[0052] Search servers may include one or more computing devices **110** designed to implement a search engine, such as a documents/records search engine, general webpage search engine, etc. Search servers may, for example, may include one or more web servers to receive search queries and/or inputs from users **105**, search one or more databases **125** in response to the search queries and/or inputs, and provide documents or information, relevant to the search queries and/or inputs, to users **105**. In some implementations, search servers may include a web search server that may

provide webpages to users **105**, where a provided webpage may include a reference to a web server at which the desired information and/or links is located. The references, to the web server at which the desired information is located, may be included in a frame and/or text box, or as a link to the desired information/document.

[0053] Document indexing servers may include one or more computing devices **110** designed to index documents available through networks. Document indexing servers may access other servers **120**, such as web servers that host content **415**, **435**, **455**, to index the content **415**, **435**, **455**. In some implementations, document indexing servers may index documents/records stored by other servers **120** connected to the network. Document indexing servers may, for example, store and index content **415**, **435**, **455**, information, and documents relating to user profiles **135** and user-generated content **415**, **435**, **455**. Web servers may include servers **120** that provide webpages to clients. For instance, the webpages may be HTML-based webpages. A web server may host one or more websites. A website, as the term is used herein, may refer to a collection of related webpages. Frequently, a website may be associated with a single domain name, although some websites may potentially encompass more than one domain name. The concepts described herein may be applied on a per-website basis. Alternatively, in some implementations, the concepts described herein may be applied on a per-webpage basis.

[0054] In some preferred embodiments, the system **100** may convert a combination of at least one digital code **114**, environmental data **135B**, and user data **135A** into a digital fingerprint **136**. In some embodiments, the data within the digital fingerprint **136** is preferably encrypted in a way such that the system **100** may extract said data from said digital fingerprint **136**. The system **100** may then tag image data **135D** with the digital fingerprint **136**. Image data **135D** tagged with an encrypted digital fingerprint **136** therefore contains data that cannot easily be altered by a user **105**. In a preferred embodiment, tagged image data **135E** created by the system **100** is a digitally watermarked image and/or an enhanced metadata image. A digitally watermarked image is image data **135D** that has a watermark in the form of the digital fingerprint **136** embedded within the pixel data. Because the pixels of the image data **135D** are changed by the digital fingerprint **136**, the watermark cannot be removed by a user **105** once the digitally watermarked image has been created by the system **100**. The system **100** may then confirm an image as legitimate by verifying the watermark within the image. An enhanced metadata image contains the digital fingerprint **136** within the metadata of the image data **135D**. In a preferred embodiment, the digital fingerprint **136** of an enhanced metadata image is encrypted in a way such that a user **105** may not easily change the digital fingerprint **136** without altering the digital fingerprint **136** in a way that prevents the system **100** from verifying the image data **135D** as legitimate.

[0055] As illustrated in FIG. 4, the system **100** may also comprise a plurality of permission levels **400** that may allow a user **105** to limit what data within their user profiles **135** they share with another user **105**. This data may be collectively known as content **415**, **435**, **455**. To access the content **415**, **435**, **455** stored within the database **125**, users **105** may be required to make a request via a user interface **111**. Access to the content **415**, **435**, **455** within the database **125** may be granted or denied by the processor **115** based on verification of a requesting user's **405**, **425**, **445** permission level **400**. If the requesting user's **405**, **425**, **445** permission level **400** is sufficient, the processor **115** may provide the requesting user **405**, **425**, **445** access to content **415**, **435**, **455** stored within the database **125**. Conversely, if the requesting user's **405**, **425**, **445** permission level **400** is insufficient, the processor **115** may deny the requesting user **405**, **425**, **445** access to content **415**, **435**, **455** stored within the database **125**. In an embodiment, permission levels **400** may be based on user roles **410**, **430**, **450** and administrator roles **470**, as shown in FIG. 4. User roles **410**, **430**, **450** allow requesting users **405**, **425**, **445** to access content **415**, **435**, **455** that a user has uploaded and/or otherwise obtained through use of the system **100**. User roles **410**, **430**, **450** allow users **105** (or requesting users **405**, **425**, **445** authorized by the user **105**) to access the data tied to their own user profiles **135** within the database **125**. Administrator roles **470** allow administrators **465** to

access system **100** wide data.

[0056] In an embodiment, user roles **410, 430, 450** may be assigned to a user **105** in a way such that a requesting user **405, 425, 445** may view user profiles **135** containing measurement data **135C**, environmental data **135B**, image data **135D**, and user data **135A** via a user interface **111**. To access the data within the database **125**, a user **105** may make a user request via the user interface **111** to the processor **115**. In an embodiment, the processor **115** may grant or deny the request based on the permission level **400** associated with the requesting user **405, 425, 445**. Only users **105** having appropriate user roles **410, 430, 450** or administrator roles **470** may access the data within the user profiles **135**. For instance, as illustrated in FIG. 2, requesting user 1 **405** has permission to view user 1 content **415** and user 2 content **435** whereas requesting user 2 **425** only has permission to view user 2 content **435**. Alternatively, user content **415, 435, 455** may be restricted in a way such that a user **105** may only view a limited amount of user content **415, 435, 455**. For instance, requesting user 3 **445** may be granted a permission level **400** that only allows them to view user 3 content **455** related to the length of wildlife/marine life **102** captured but not user 3 content **455** related to the geolocations in which said wildlife/marine life **102** was captured. In the example illustrated in FIG. 2, an administrator **465** may bestow a new permission level **400** on users **105** so that it may grant them greater permissions or lesser permissions. For instance, an administrator **465** may bestow a greater permission level **400** on other users **105** so that they may view user 3's content **455** and/or any other user's content **415, 435, 455**. Therefore, the permission levels **400** of the system **100** may be assigned to users **105** in various ways without departing from the inventive subject matter described herein.

[0057] FIG. 5 provides a flow chart **500** illustrating certain, preferred method steps that may be used to carry out the method for capturing image data **135D**, extracting measurement data **135C** and a pattern from said image data **135D**, and confirming the measurement data **135C** as legitimate. Step **505** indicates the beginning of the method. During steps **510**, the processor **115** may collect image data **135D** using the camera of the computing device **110**. Once collected, the processor **115** may save the image data **135D** within a user profile **135** of the system **100** during step **515**. The processor **115** may then perform a query to determine if measurement data **135C** can be collected from the image data **135D** during step **517**. Based on the results of the query, the processor **115** may perform an action during step **520**. If the processor **115** determines that measurement data **135C** may be extracted from the image data **135D**, the processor **115** may extract the image data **135D** and subsequently save it during step **525** and **527**, respectively. If the processor **115** determines that measurement data **135C** may not be extracted from the image data **135D**, the processor **115** may alert a user **105** via the user interface **111** of an error during step **523** and subsequently proceed to the terminate method step **560**.

[0058] Once the measurement data **135C** has been extracted, the processor **115** may perform a query to determine whether or not a pattern may be extracted from the image data **135D** during step **530**. Based on the results of the query, the processor **115** may take an action during step **535**. If the processor **115** determines that no pattern may be extracted from the image data **135D**, the processor **115** may proceed to step **557** and tag the image data **135D** and measurement data **135C** as illegitimate. If the processor **115** determines that a pattern may be extracted from the image data **135D**, the processor **115** may extract the pattern during step **540**. Once the pattern has been extracted, the processor **115** may transform the pattern into a digital code **114** during step **545**. The system **100** may use a cypher to transform the pattern into a digital code **114**, wherein a cypher may be defined as a set of rules used to convert a pattern of numbers and/or letters into a seemingly different set of numbers and/or letters. The system **100** may then perform a query to determine if the digital code **114** matches a digital key of the system **100** during step **550**. The processor **115** may perform an action based on the results of the query during step **555**. If the processor **115** determines that the digital code **114** does not match the digital key, the system **100** may tag the data as illegitimate during step **557**. If the system **100** determines that the digital code **114** does match

the digital key, the system **100** may tag the data as legitimate during step **558**. Once the system **100** has determined that the data is legitimate or illegitimate, the system **100** may proceed to terminate method step **560**.

[0059] FIG. **6** provides a flow chart **600** illustrating certain, preferred method steps that may be used to carry out the method of embedding a watermark containing encrypted environmental data **135B**, image data **135D**, and/or measurement data **135C** on an image captured by a camera of a computing device **110**. Step **605** indicates the beginning of the method. During steps **610**, the processor **115** may perform a query to determine whether or not a digital code **114** may be recognized within image data **135D** collected by the camera, wherein the recognized digital code **114** is one that matches a digital lock **113** of the system **100**. Based on the results of the query, the processor **115** may take an action during step **615**. If the processor **115** determines that there is no recognizable digital code **114**, the processor **115** may proceed to the terminate method step **655**. If the processor **115** determines that there is a recognizable digital code **114**, the processor **115** may activate the UI of the computing device **110** in a way that may allow a user **105** to capture and save image data **135D** with the camera during step **620**. The computing device **110** may then capture image data **135D** based on an input of the user **105** during step **625**. In some preferred embodiments, the digital code **114** may be received by the computing device **110** from a wireless communication device of the control board **210**, which may then be compared to a digital lock **113** of the system **100** prior to granting the user **105** access to the camera to capture image data **135D**.

[0060] Once the image data **135D** has been captured, the system **100** may save the image data **135D** within the non-transitory computer-readable medium **116** and/or database **125** during step **630**. The system **100** may then save environmental data **135B**, measurement data **135C**, and/or pattern associated with said image data **135D** during step **635**. In a preferred embodiment, geolocation data and length are associated with image data **135D** by the system **100**. The processor **115** may then create a digital fingerprint **136** containing the image data **135D**, environmental data **135B**, measurement data **135C**, and/or pattern during step **640**. In a preferred embodiment, a digital fingerprint **136** is created using a cryptographic hash function **130**. A cryptographic hash function **130** may be defined as a mathematical algorithm that maps data of arbitrary size to a bit array of a fixed size. A cryptographic hash function **130** is preferably a one-way function, that is, a function which is practically infeasible to invert. Once the digital fingerprint **136** has been created, the system **100** may then create a digital image from the image data **135D** during step **642** and subsequently place the digital fingerprint **136** over the digital image to create a digitally watermarked image during step **645**. The digitally watermarked image may then be saved within the non-transitory computer-readable medium **116** and/or database **125** by the processor **115** during step **650**. The system **100** may then proceed to the terminate method step **655**.

[0061] The subject matter described herein may be embodied in systems, apparatus, methods, and/or articles depending on the desired configuration. In particular, various implementations of the subject matter described herein may be realized in digital electronic circuitry, integrated circuitry, specially designed application specific integrated circuits (ASICs), computer hardware, firmware, software, and/or combinations thereof. These various implementations may include implementation in one or more computer programs that may be executable and/or interpretable on a programmable system including at least one programmable processor, which may be special or general purpose, coupled to receive data and instructions from, and to transmit data and instructions to, a storage system, and at least one input/output device.

[0062] These computer programs, which may also be referred to as programs, software, software applications, applications, components, or code, may include machine instructions for a programmable processor, and may be implemented in a high-level procedural and/or object-oriented programming language, and/or in assembly machine language. As used herein, the term “non-transitory computer-readable medium” refers to any computer program, product, apparatus, and/or device, such as magnetic discs, optical disks, memory, and Programmable Logic Devices

(PLDs), used to provide machine instructions and/or data to a programmable processor, including a non-transitory computer-readable medium that receives machine instructions as a computer-readable signal. The term “computer-readable signal” refers to any signal used to provide machine instructions and/or data to a programmable processor. To provide for interaction with a user, the subject matter described herein may be implemented on a computer having a display, such as a cathode ray tube (CRT), liquid crystal display (LCD), light emitting display (LED) monitor for displaying information to the user and a keyboard and a pointing device, such as a mouse or a trackball, by which the user may provide input to the computer. Displays may include, but are not limited to, visual, auditory, cutaneous, kinesthetic, olfactory, and gustatory displays, or any combination thereof.

[0063] Other kinds of devices may be used to facilitate interaction with a user as well. For instance, feedback provided to the user may be any form of sensory feedback, such as visual feedback, auditory feedback, or tactile feedback; and input from the user may be received in any form including, but not limited to, acoustic, speech, or tactile input. The subject matter described herein may be implemented in a computing system that includes a back-end component, such as a data server, or that includes a middleware component, such as an application server, or that includes a front-end component, such as a client computer having a graphical user interface or a Web browser through which a user may interact with the system described herein, or any combination of such back-end, middleware, or front-end components. The components of the system may be interconnected by any form or medium of digital data communication, such as a communication network. Examples of communication networks may include, but are not limited to, a local area network (“LAN”), a wide area network (“WAN”), metropolitan area networks (“MAN”), and the internet.

[0064] The implementations set forth in the foregoing description do not represent all implementations consistent with the subject matter described herein. Instead, they are merely some examples consistent with aspects related to the described subject matter. Although a few variations have been described in detail above, other modifications or additions are possible. In particular, further features and/or variations may be provided in addition to those set forth herein. For example, the implementations described above may be directed to various combinations and subcombinations of the disclosed features and/or combinations and subcombinations of several further features disclosed above. In addition, the logic flow depicted in the accompanying figures and/or described herein do not necessarily require the particular order shown, or sequential order, to achieve desirable results. It will be readily understood to those skilled in the art that various other changes in the details, materials, and arrangements of the parts and method stages which have been described and illustrated in order to explain the nature of this inventive subject matter may be made without departing from the principles and scope of the inventive subject matter.

Claims

1. A system for confirming image data comprising: a light configured to emit a pattern, a camera configured to capture image data containing said pattern, a computing device operably connected to said camera and having a user interface configured to allow manipulation of said camera, wherein said computing device contains user data pertaining to a user of said computing device, and a processor operably connected to said computing device and said camera, wherein said processor extracts said pattern from said image data, wherein said processor creates tagged image data using said image data, user data, and said pattern.
2. The system of claim 1, wherein said pattern is specific to a particular event.
3. The system of claim 2, wherein said tagged image data is associated with said particular event.
4. The system of claim 1, further comprising a measuring device configured to measure attributes of at least one of marine life or wildlife.

5. The system of claim 4, wherein said processor extracts measurement data from said image data containing said measuring device and said at least one of marine life or wildlife.
 6. The system of claim 5, wherein said processor creates said tagged image data using said user data, pattern, and measurement data.
 7. The system of claim 1, further comprising a user identification (UID) pattern, wherein said UID pattern is associated with said user.
 8. The system of claim 7, wherein said processor extracts said UID pattern from said image data containing said UID pattern.
 9. The system of claim 8, wherein said processor creates said tagged image data using said user data, pattern, and UID pattern.
 10. The system of claim 1, further comprising at least one sensor operably connected to said processor and configured to obtain environmental data.
 11. The system of claim 10, wherein said processor creates said tagged image data using said user data, pattern, and environmental data.
 12. The system of claim 1, further comprising one of hunting equipment, fishing equipment, or card on which said light is secured.
 13. A system for confirming image data comprising: a measuring device configured to measure physical attributes of at least one of marine life or wildlife, a light secured to said measuring device and configured to emit a pattern, a computing device having a camera and a user interface configured to manipulate said camera, wherein said camera is configured to capture image data containing said pattern, at least one of marine life or wildlife, and measuring device, and a processor operably connected to said computing device and said camera, wherein said processor extracts said pattern from said image data containing said light, wherein said processor extracts measurement data from said image data pertaining to said physical attributes of said at least one of marine life or wildlife, wherein said processor creates tagged image data using said image data, pattern, and said measurement data.
 14. The system of claim 13, further comprising a user identification (UID) pattern, wherein said UID pattern is associated with a user.
 15. The system of claim 14, wherein said processor extracts said UID pattern from said image data containing said UID pattern.
 16. The system of claim 15, wherein said processor creates said tagged image data using said measurement data, pattern, and UID pattern.
 17. The system of claim 13, further comprising at least one sensor operably connected to said processor and configured to obtain environmental data.
 18. The system of claim 17, wherein said processor creates said tagged image data using said measurement data, pattern, and environmental data.
 19. The system of claim 13, further comprising one of hunting equipment, fishing equipment, or card on which a UID pattern is secured, wherein said UID pattern is associated with a user.
 20. A system for confirming image data comprising: a non-transitory computer-readable medium coupled to a processor and having instructions stored thereon, which, when executed by said processor, cause said processor to perform operations comprising: receiving image data from a camera operably connected to a computing device, wherein said image data contains a light and at least one of marine life or wildlife, receiving user data from said computing device, wherein said user data instructs said processor which user of a plurality of users captured said image data is associated, extracting a pattern emitted by said light from said image data, and creating tagged image data using said image data, pattern, and said user data.
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